

Project Proposal

Knowledge Graphs

25/26

Important Dates

Deliverable	Weight	Due Date
D1	0%	13.04.2026
D2	10%	27.04.2026
D3	25%	25.05.2026
Poster	15%	03.06.2026

Overview

One of the main goals of this course is to apply **knowledge graph methods** to real-world problems or research questions. In this project, you will use the concepts and tools learned during the semester—such as graph embeddings, link prediction, entity resolution, reasoning, and semantic querying—to tackle a problem of your choice.

The topic is open, and you may choose one aligned with your personal interests, MSc specialization, or potential thesis work. Your project can focus on any type of structured or semi-structured data (e.g., RDF datasets, biomedical ontologies, or knowledge bases like Wikidata or DBpedia).

Keep in mind computational feasibility; Google Colab, Kaggle, or other cloud platforms can be used. Some GPU access may be available in Room 1.3.30 if needed for graph neural network experiments.

Project Stages

1. Deliverable 1: Topic Proposal

Prepare a 5-minute presentation including:

- **Title & group members**
- **Problem** – What KG problem will you study (e.g., link prediction, entity alignment, reasoning, querying, modelling of a specific domain, etc.) and why it is interesting.
- **Background sources** – Papers, datasets, or existing benchmarks.
- **Data** – Which knowledge graph(s) will be used; if new data is collected, how.
- **Method/Algorithm** – examples: Graph embeddings, GNNs, rule-based reasoning, or hybrid approaches; note any existing implementations you may leverage.
- **Evaluation** – Metrics like Hits@K, MRR, precision/recall, or qualitative analysis via visualization of subgraphs.

2. Deliverable 2: Report

Write the first part of the report:

- Introduction: problem, motivation, and overall approach
- Problem statement: dataset, evaluation plan, expected results
- Methodology: algorithms, architecture, tasks (with a methodology figure)
- Planned experiments & preliminary results
- **Maximum 4 pages**

Template: <https://github.com/cvpr-org/author-kit/releases>

3. Deliverable 3: Report + Code

Complete the report:

- Training methods – extended experiments with embeddings, GNNs, reasoning modules.
- Experiments & results – evaluation metrics, learning curves, qualitative visualizations.
- Contribution of each member – tasks completed.
- Include tables, figures, metrics, and any updated methodology.
- **Maximum 8 pages** (without references)
- Submit the report (PDF) and the jupyter notebook used to develop the project

4. Poster

- A1 portrait-oriented, colored poster.
- Present your problem, methodology, experiments, and results.
- Example layout: problem → data → method → results → conclusion.
- Poster session: Atrium of C6, June 3rd, 16:30.
- Poster template and printing info same as DL course.
- Submit poster (PDF)

Submission & Groups

- Groups: max 3/4 students; solo projects not allowed except for working students or other special status.

Resources

Knowledge Graph Datasets

- **Wikidata** – Massive multilingual open knowledge graph (structured RDF triples).
<https://www.wikidata.org/>
- **DBpedia** – Structured data extracted from Wikipedia in RDF.
<https://www.dbpedia.org/>
- **YAGO** – Semantic knowledge base combining data from Wikidata/Wikipedia/WordNet.
<https://yago-knowledge.org/> ([Wikipedia](#))
- **OpenBioLink** – Biomedical link prediction benchmark and KG resource.
<https://github.com/OpenBioLink/OpenBioLink>
- **Other biomedical KGs** and ontologies– e.g., PrimeKG, Hetionet, Bioportal

Benchmarks for Link Prediction / Graph Completion

- **FB15k-237** – Standard KG embedding benchmark derived from Freebase (no inverse relations).
Available in many frameworks and repositories (e.g., PyKEEN datasets).
(pykeen.readthedocs.io)
- **WN18RR** – WordNet subset benchmark with inverse relations removed.
Included as a core evaluation dataset for link prediction tasks.
(pykeen.readthedocs.io)
- **OGBL** – Suite of benchmarks from the Open Graph Benchmark (OGB) project for large-scale graph tasks.
See <https://ogb.stanford.edu/> (OGB datasets)

(Many of these are accessible directly via libraries like PyKEEN.) (pykeen.readthedocs.io)

Key Conferences and Paper Sources

- **International Semantic Web Conference (ISWC)** – semantic web & knowledge graph research venue.
<https://iswc2026.semanticweb.org/>
- **Extended Semantic Web Conference (ESWC)** – research on semantic technologies and KGs.
<https://eswc-conferences.org/>
- **ACM SIGKDD Conference on Knowledge Discovery and Data Mining** – ML and KG reasoning applications.
<https://kdd.org/>
- **NeurIPS Graph Representation Learning Workshops** – state-of-the-art embedding methods & evaluations.
<https://neurips.cc/Conferences/2025/WorkshopTrack>
- **ACL/SIGDAT (Computational Linguistics)** – papers on KG in natural language tasks.
<https://www.aclweb.org/portal/>

(Use these as source points for literature reviews, benchmarks, or method inspiration.)

Tools and Libraries

- **PyKEEN** – Python framework for training and evaluating knowledge graph embedding models.
<https://pykeen.readthedocs.io/> (pykeen.readthedocs.io)
- **DGL-KE** – Distributed Knowledge Embedding library for large graphs (part of DGL ecosystem).
<https://www.dgl.ai/>
- **RDFlib** – Python library for working with RDF and SPARQL.
<https://rdflib.dev/>
- **GraphDB - a graph database with native support for RDF**
<https://graphdb.ontotext.com/>